5. Missionaries and Cannibals Problem

Aim

To move three missionaries and three cannibals across a river without allowing cannibals to outnumber missionaries on either bank when missionaries are present.

Algorithm

1. Represent a state as (missionaries on left, cannibals on left, boat side).
2. Start from (3, 3, 0); the goal is (0, 0, 1).
3. Generate boat moves carrying one or two people.
4. Check that the numbers stay within 0 to 3 and missionaries are never outnumbered on either bank unless none are present.
5. Use BFS and store parent states to avoid revisiting states.
6. Print the path if the goal is reached.

Program

AI.py - C:/Users/Barath/AppData/Local/Programs/Python/Python313/AI.py (3.13.15)

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```
from collections import deque

start = (3, 3, 0)
goal = (0, 0, 1)
moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]

def safe(m, c):
    return 0 <= m <= 3 and 0 <= c <= 3 and (m == 0 or m >= c)

q = deque([start])
parent = {start: None}

while q:
    state = q.popleft()
    if state == goal:
        break
    m, c, boat = state

    for dm, dc in moves:
        if boat == 0:
            nxt = (m-dm, c-dc, 1)
        else:
            nxt = (m+dm, c+dc, 0)

        ml, cl, _ = nxt
        mr, cr = 3-ml, 3-cl

        if (safe(ml, cl) and safe(mr, cr)
                and nxt not in parent):
            parent[nxt] = state
            q.append(nxt)

if goal in parent:
    path = []
    s = goal
    while s is not None:
        path.append(s)
        s = parent[s]
    for s in reversed(path):
        print(s)
else:
    print("No solution")
```

Output

```
>
========== RESTART: C:/Users/Barath/AppData/Local/Programs/Python/Python313/AI.py ==========
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)
>
```

Result

Thus, all missionaries and cannibals were transported safely to the opposite bank.